



Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Mini project Instructions

Course	Computer Vision and Image Processing
Course Code	EE7204 / EC7205
Duration	12 weeks
Outcomes	The aim of the mini project is to, • Apply theoretical knowledge of image processing and computer vision to a practical problem. • Encourage creativity and research-oriented problem solving. • Develop basic implementation and reporting skills.

Project Requirement

The group must propose a Mini Project that primarily focuses on Image Processing and Computer Vision techniques.

The project must consist of **at least 65% image processing and computer vision components.**

Example Project Areas

Students may select topics from, but not limited to, the following areas,

- Image segmentation and object detection
- Image enhancement or restoration
- Feature extraction and pattern recognition
- Medical image analysis
- Face, gesture, or motion detection
- Image classification or recognition using deep learning
- Optical character recognition (OCR)
- Video-based analysis or tracking
- Image-based measurement systems (e.g., counting, defect detection, etc.)

Group Selection Criteria

- Each group can have **up to 4 members.**
- Students may **freely select their own group members.**
- Groups can include members from **both Electrical and Computer Engineering departments.**
- Once selected, the same group will be **considered for future course group work** as well.
- After finalizing your group members, **fill out the Google Sheet shared with you using the provided link,**
- **When you update the Google Sheet, merge the columns for *Group Name*, *Project Title*, *Group No.*, and *Proposal Acceptance Status* before submission.**

Note: Groups must be finalized **before submitting the project proposal**. The deadline for selecting the group will be **31st of December**.

Proposal Format

1. Each group must prepare and submit a **Mini Project Proposal** using **only the LaTeX template provided**.
 - The template already includes all required sections and formatting.
 - **Do not modify** the structure, margins, or layout.
 - Don't remove the "back to table content" function from the LaTeX
 - **Words documents are not allowed.**
2. The proposal must focus on **Image Processing and Computer Vision** concepts.
 - At least **65% of the project content** must involve image processing or computer vision-based components.
 - AI and plagiarism content must be less than 20%.
3. You may include **sketches, diagrams, or figures** if necessary to explain your idea clearly. (if you use sketches, diagrams, or figures or table, add table of contents for each one)
4. **Follow the word limits** indicated in each section of the LaTeX template strictly.
 - Exceeding or falling far below the limit will result in **mark deductions**.
 - Each section has a recommended range.
5. The **clickable Table of Contents** in the template must remain functional.
6. **Proposal length:** The complete proposal should be **6-20 pages** (including figures and table).
7. Use clear academic writing and avoid plagiarism.
8. Save your final proposal as a **PDF file** generated from the LaTeX source.
9. Submit the **PDF version** of the proposal.

Submission date will be **14th of January**. **Submission box will be given by your ELMS.**

Rename the PDF file with your group number before submitting the report to ELMS.